Overfitting

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable capacity-comparison record.

Trace training leaves,validation predictions and the limits of gap-based diagnosis.

Keep one habit: W1–W3/W5 → selected new accuracy task → project capacity record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Reproducible Experiments

> **My remembered idea, example and correction**

Train, Validation, Test

> **My remembered idea, example and correction**

Absolute Error and MAE

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main supplied MAEs:constanttrain 5/validation 6, line 2/3, flexiblecurve 0/8; choose line by smallest validation MAE. Picture Small85%/80%gap 5 points, Large100%/60%gap 40; revisedLargevalidation82%givesgap 18 and highest pictured validation. Separate paper-tree trainingT1(0, 0 clear), T2(0, 1 clear), T3(1, 0 blocked), T4(1, 1 clear); validationV1/V2 clear, V3(1, 0 clear), V4(1, 1 blocked).

  Near-first/tall-second, clearleaf ties:depth 0/1 train3/4 valid3/4; depth 2 train4/4 valid2/4. Selectdepth0by declared shallower tie. Gap alone does not prove cause.

- Reference for this exercise: open this week's student lesson, "Overfitting", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Choose among the three worked candidates and explain the **training** trap.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main supplied MAEs:constanttrain 5/validation 6, line 2/3, flexiblecurve 0/8; choose line by smallest validation MAE. Picture Small85%/80%gap 5 points, Large100%/60%gap 40; revisedLargevalidation82%givesgap 18 and highest pictured validation. Separate paper-tree trainingT1(0, 0 clear), T2(0, 1 clear), T3(1, 0 blocked), T4(1, 1 clear); validationV1/V2 clear, V3(1, 0 clear), V4(1, 1 blocked).

  Near-first/tall-second, clearleaf ties:depth 0/1 train3/4 valid3/4; depth 2 train4/4 valid2/4. Selectdepth0by declared shallower tie. Gap alone does not prove cause.

- Does a **training**-validation gap alone prove **overfitting**?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main supplied MAEs:constanttrain 5/validation 6, line 2/3, flexiblecurve 0/8; choose line by smallest validation MAE. Picture Small85%/80%gap 5 points, Large100%/60%gap 40; revisedLargevalidation82%givesgap 18 and highest pictured validation. Separate paper-tree trainingT1(0, 0 clear), T2(0, 1 clear), T3(1, 0 blocked), T4(1, 1 clear); validationV1/V2 clear, V3(1, 0 clear), V4(1, 1 blocked).

  Near-first/tall-second, clearleaf ties:depth 0/1 train3/4 valid3/4; depth 2 train4/4 valid2/4. Selectdepth0by declared shallower tie. Gap alone does not prove cause.

- Describe one fair next development comparison.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main supplied MAEs:constanttrain 5/validation 6, line 2/3, flexiblecurve 0/8; choose line by smallest validation MAE. Picture Small85%/80%gap 5 points, Large100%/60%gap 40; revisedLargevalidation82%givesgap 18 and highest pictured validation. Separate paper-tree trainingT1(0, 0 clear), T2(0, 1 clear), T3(1, 0 blocked), T4(1, 1 clear); validationV1/V2 clear, V3(1, 0 clear), V4(1, 1 blocked).

  Near-first/tall-second, clearleaf ties:depth 0/1 train3/4 valid3/4; depth 2 train4/4 valid2/4. Selectdepth0by declared shallower tie. Gap alone does not prove cause.

- What extra **evidence** would you inspect before concluding a model memorized noise?

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the selected A:training 96%, validation 62%; B:training 88%, validation 79%, on the stated same valid evaluation conditions. Compute training-minus-validation accuracy gaps in percentage points, then choose by validation accuracy. Do not substitute the pictured 85/80or100/60 results or the paper-tree denominators. No causal diagnosis or untouched final result follows from this aggregate comparison alone.

- Model A **training**96%, validation62%; model B **training**88%, validation79%. Compare gaps and choose a development candidate, assuming the same valid evaluation conditions.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Check Plant Classifier train-validation gaps and limit **complexity** before final test release.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Interpret a gap for **Overfitting** and model **capacity**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A has **training** 98%, validation 58%; B has **training** 86%, validation 81%. Compute gaps and select a candidate with a qualification.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A scores100% train,**60%** validation; B scores85%,**80%** on the same sets.
- Compute gaps and select by validation accuracy.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Model A has **training** accuracy 99% and validation accuracy **70%**; B has 88% and 84%. Compute each gap and discuss the **evidence**.
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 27 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A has **training** 98%, validation 58%; B has **training** 86%, validation 81%. Compute gaps and select a candidate with a qualification.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

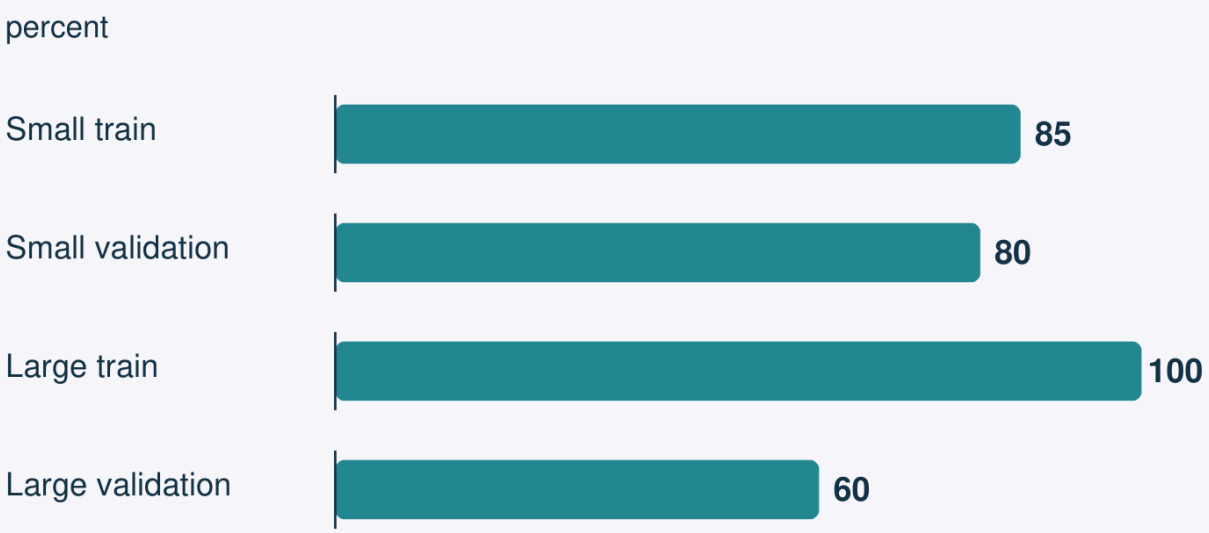
Task and evidence record

- See **overfitting** as a gap requiring investigation, not a label based only on model size.
- Toy model Small: **training** 85%, validation **80%**. Large: **training** 100%, validation **60%**. Same split and scoring rule.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Does100% **training** accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/27

Training and validation accuracy



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/27

Overfitting

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gaps

Small gap=5 percentage points; Large gap=40 percentage points.



2. Select development candidate

Small has the higher validation accuracy despite its lower training fit.



3. Investigate large model

Check capacity, training choices and data mismatch without tuning on final cases.

CHANGE ONE THING

Large's validation accuracy rises to82% after a development revision.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project portfolio milestone

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- See **overfitting** as a gap requiring investigation, not a label based only on model size.
- Which labelled step matches this detail? Check **capacity**, **training** choices and data mismatch without tuning on final cases.
- 1. Select development candidate
- 2. Compute gaps
- 3. Investigate large model
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Investigate large model Check **capacity**, **training** choices and data mismatch without tuning on final cases. Read the picture in order.

Check **capacity**, **training** choices and data mismatch without tuning on final cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Large's validation accuracy rises to82% after a development revision. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Its gap is 18 percentage points; it now beats Small on this validation set.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Its gap is 18 percentage points; it now beats Small on this validation set.

Validation performance improved, but the **training**-validation gap is not a complete release rule.

Repair and keep the record

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as4.67 cm misleading?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

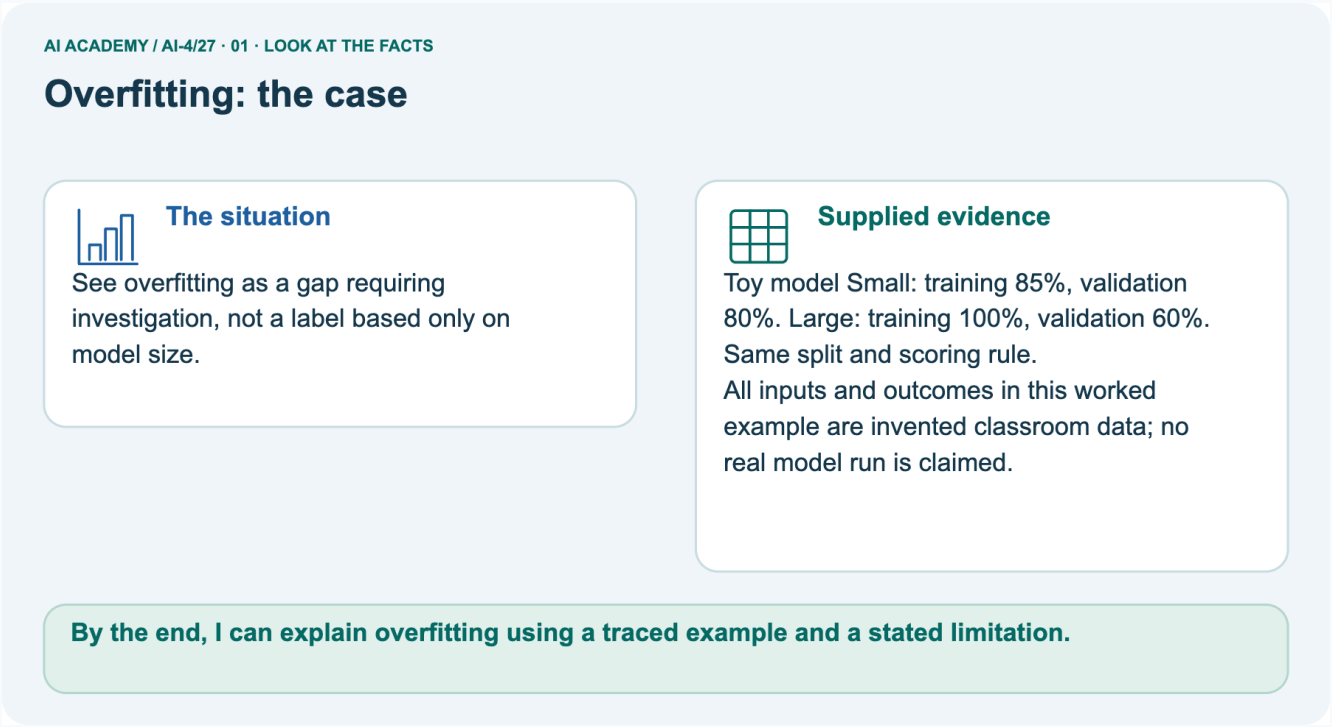
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Overfitting: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

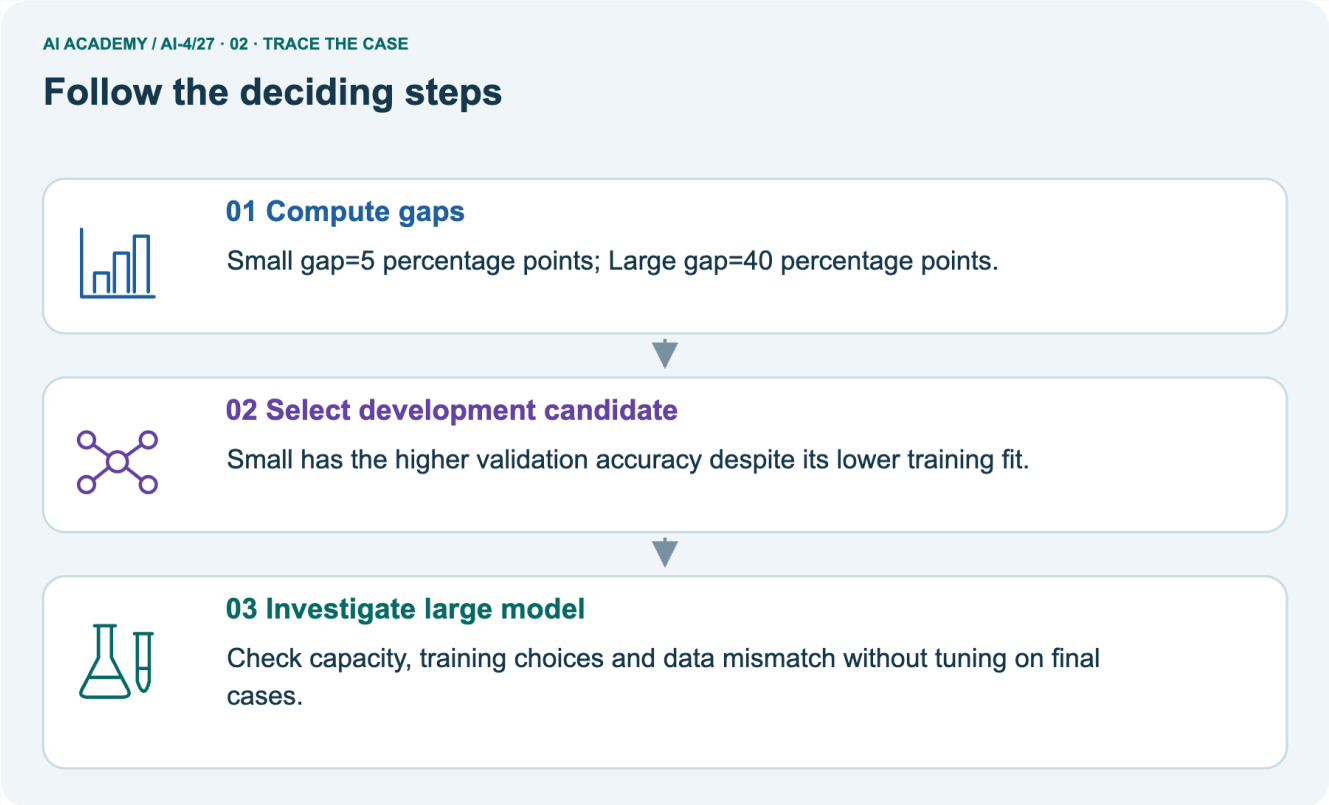
See **overfitting** as a gap requiring investigation, not a label based only on model size. Supplied facts: Toy model Small: **training** 85%, validation **80%**. Large: **training** 100%, validation **60%**. Same split and scoring rule.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



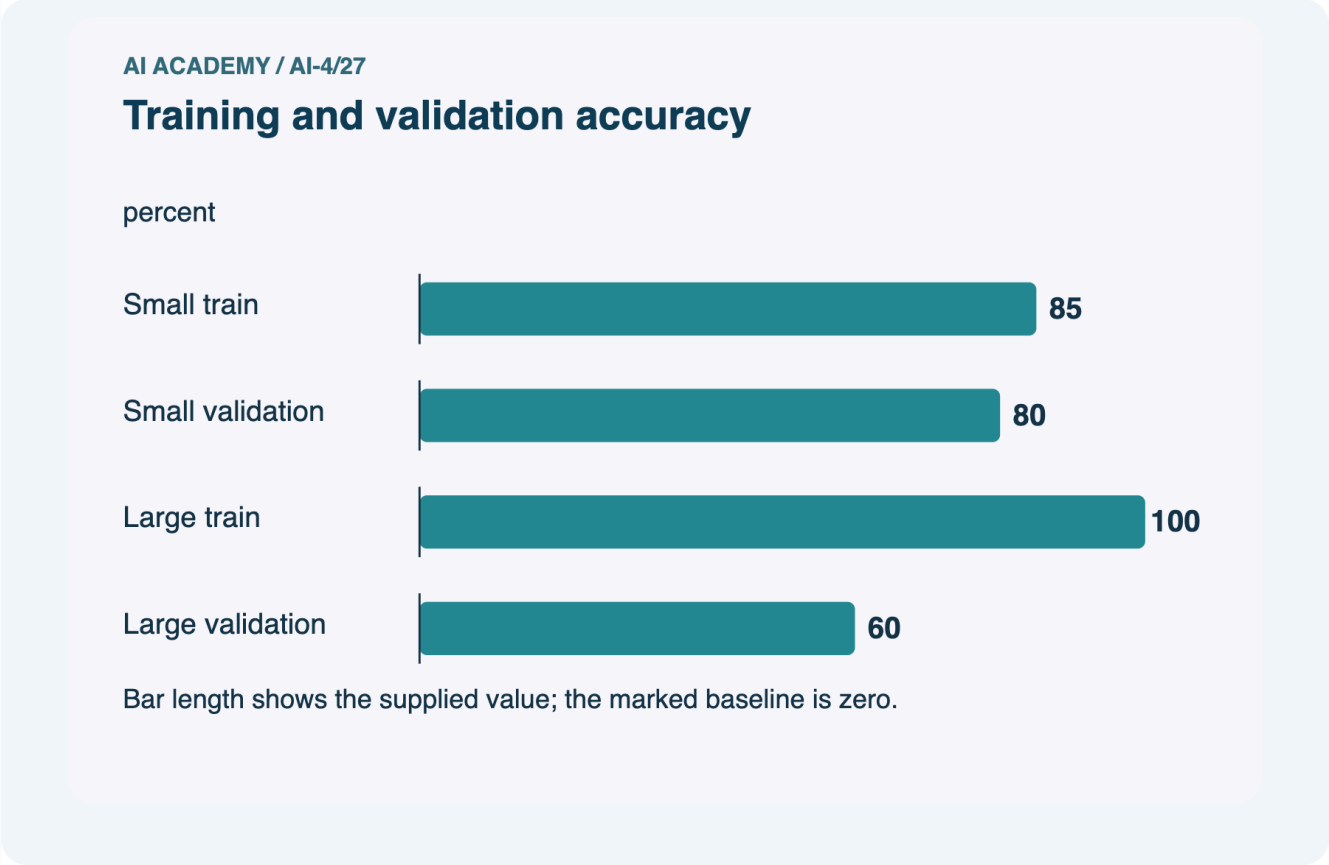
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute gaps: Small gap=5 percentage points; Large gap=40 percentage points. → Select development candidate: Small has the higher validation accuracy despite its lower **training** fit. → Investigate large model: Check **capacity**, **training** choices and data mismatch without tuning on final cases.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Training and validation accuracy



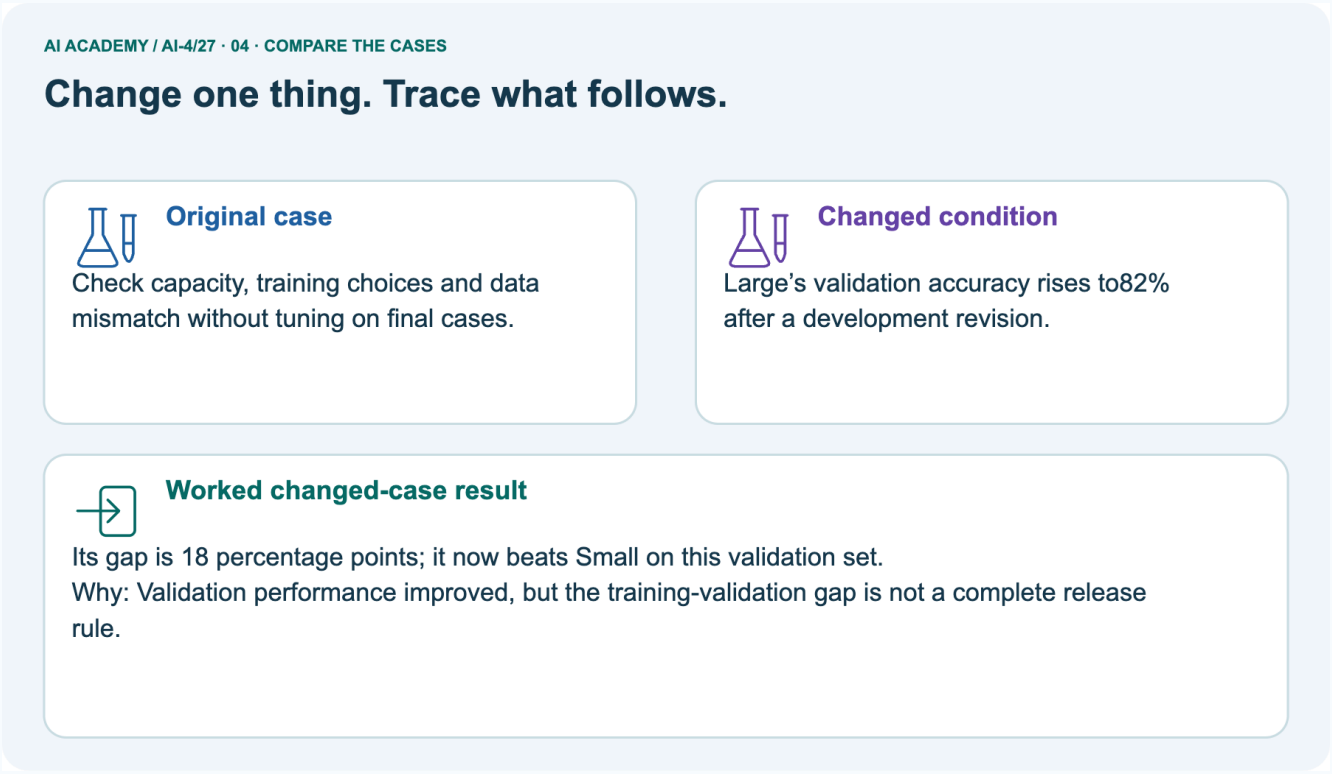
Follow the labelled picture with the matching explanation. Open the vector diagram

Training and validation accuracy. Units: percent. Small train: 85. Small validation: 80. Large train: 100. Large validation: 60.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

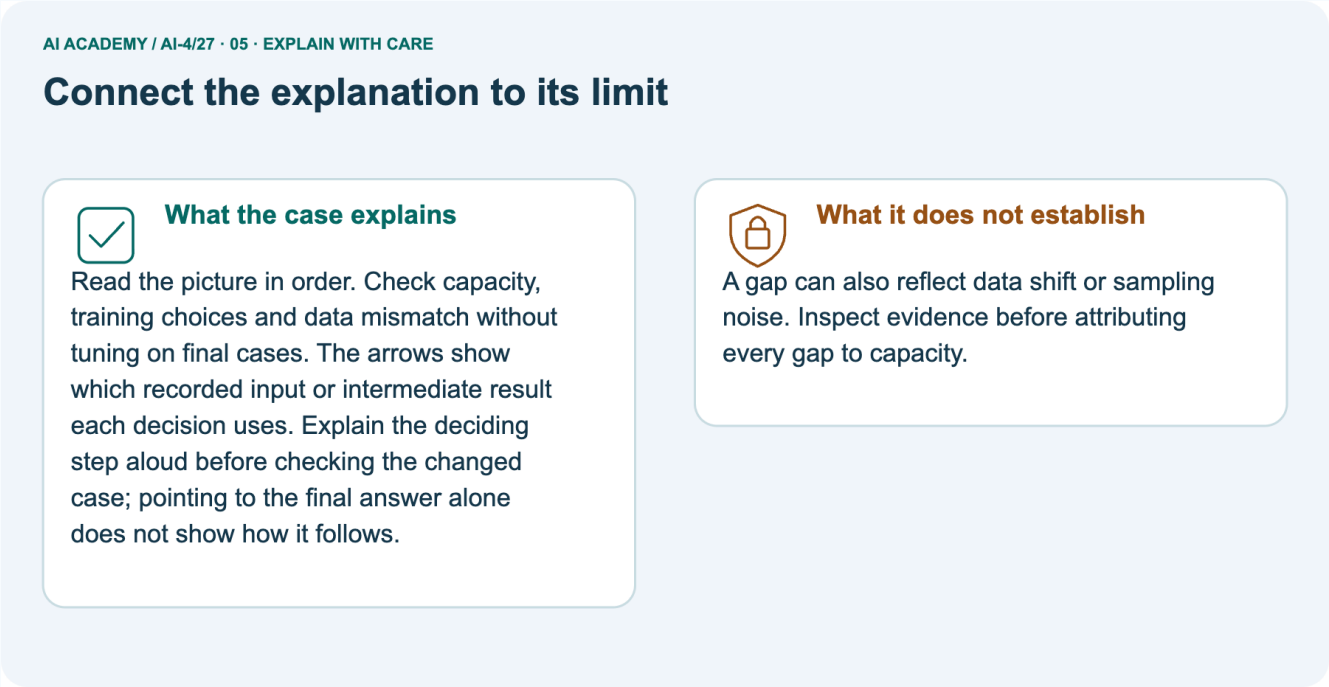
Original case: Check **capacity**, **training** choices and data mismatch without tuning on final cases. Changed condition: Large's validation accuracy rises to82% after a development revision.

Expected result: Its gap is 18 percentage points; it now beats Small on this validation set. Reason: Validation performance improved, but the **training**-validation gap is not a complete release rule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Check **capacity**, **training** choices and data mismatch without tuning on final cases. The arrows show which recorded **input** or intermediate result each decision uses.

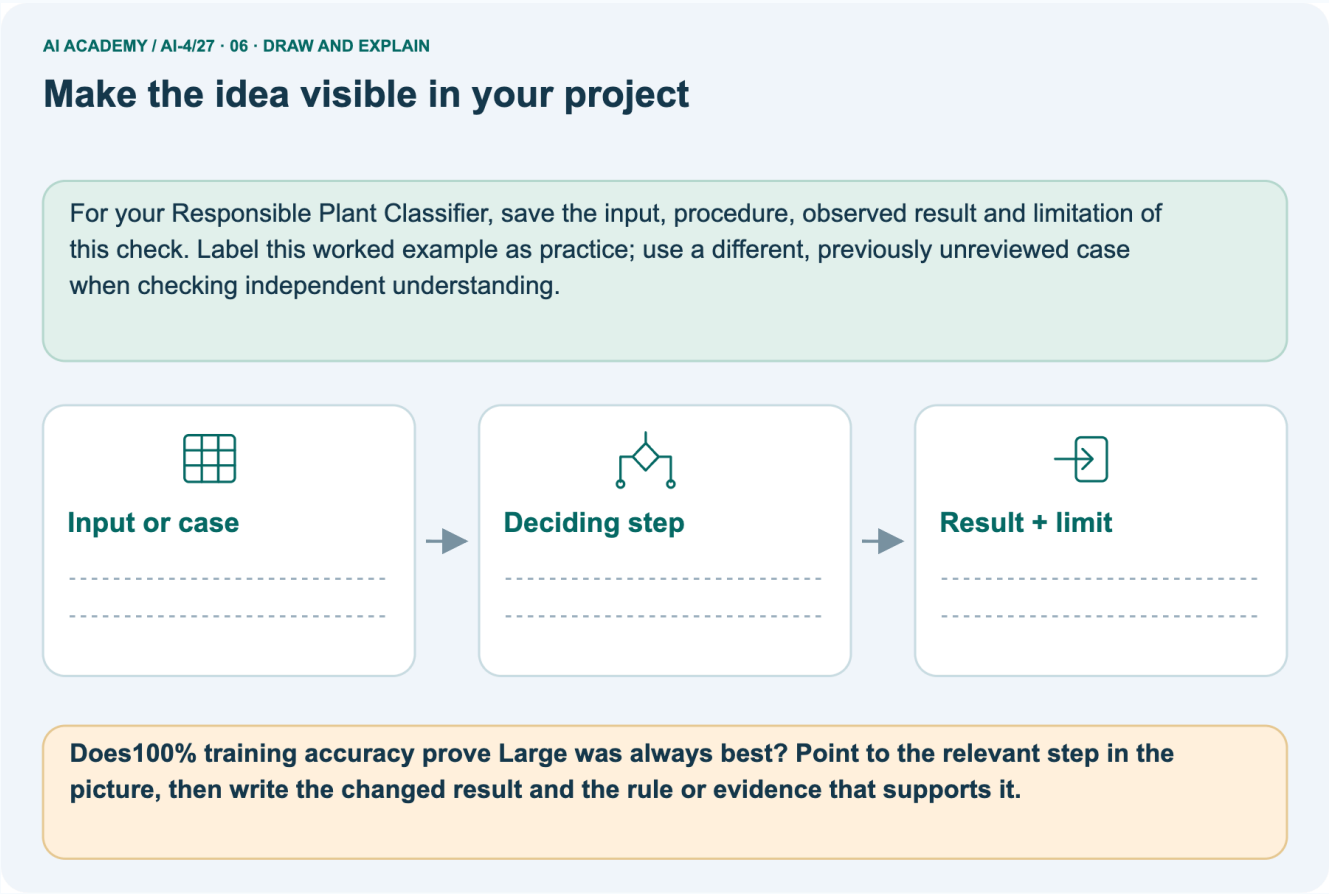
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: A gap can also reflect data shift or sampling noise.

Inspect **evidence** before attributing every gap to **capacity**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does100% **training** accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Overfitting: the case

02. Follow the deciding steps

03. Training and validation accuracy

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.